AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-07-30 | Entry window 30 July 2026 | 18 records in scope

18 RECORDS IN SCOPE (9 SUBTOPICS)	**10** EFFECT ESTIMATES EXTRACTED	**6** RECORDS FROM CHINA (33% OF CORPUS)	**4** SOUTH ASIA RECORDS (2 PREPRINTS)	**1** SENSING / INSTRUMENTATION

Signal of the day

Aerosol liquid water content is now available as a 1 km daily gridded field for the contiguous US — the missing thermodynamic covariate in both sensor calibration and toxicity-weighted exposure.

Zhang et al. (*npj Clim Atmos Sci*) trained machine-learning surrogates on GEOS-Chem output to reproduce the thermodynamic relationship between ALWC and its predictors, then applied those relationships to bias-corrected high-resolution inputs, yielding a **1 km $\times$ 1 km daily ALWC field for 2000–2019**. The ML product resolves daily variation and spatial heterogeneity that the parent CTM does not; ALWC is highest in the Midwest and lowest in the West, and **declined across most regions over the period, driven primarily by sulfate reduction**. The authors then show ALWC constrains variability in water-soluble iron, a health-relevant redox-active metal fraction.

For distributed low-cost PM sensing this matters twice over. First, ALWC is the physical quantity that hygroscopic-growth corrections are proxying for when they regress on relative humidity; a gridded ALWC field is a strictly better covariate than RH alone, because it already carries the composition dependence that makes RH corrections non-transferable between regions. Second, the sulfate-driven secular decline means a correction function fitted on a 2015 co-location is being applied to an aerosol whose water content has since shifted — a mechanism for calibration drift that is compositional rather than electronic, and which no amount of sensor-side quality control will detect.

What changed today

- **The instrumentation axis returned almost nothing this window — and that is a measurement about the pipeline, not the field.** PubMed does not index *Atmos. Meas. Tech.* or *Atmos. Chem. Phys.*, and the Consensus sweep that exists to close that gap has *no entry-date filter*: every low-cost-sensor paper it surfaced resolved to a January–July 2026 publication date, and one (Ginsburg et al., *Aerosol Sci Technol*) was already carried in the 29 July issue. All eight were rejected rather than passed off as new. See *Provenance*.
- **Source apportionment reached the shipping sector with isotopic evidence.** Chen et al. attribute **17.6–37.9% of $PM_{2.5}$ nitrate** across five Chinese coastal cities to shipping, using a dual-isotope Bayesian mixing model on 615 daily samples — with the striking regional inversion that shipping contribution *rises* with pollution severity in northern cities and *falls* in southern ones. Shipping NO_x standards remain looser than road-vehicle standards; this is the quantitative case against that asymmetry.
- **A clean negative-direction effect estimate in the reproductive literature.** Li et al. report full-cycle O_3 associated with a **15.45% reduction in sperm concentration (95% CI $-$20.95 to $-$9.58)**, sub-staged across the meiotic window with the peak at days 31–45. The design — quarterly panel over a full 90-day spermatogenic cycle in 18–22-year-olds — is unusually well matched to the biology. The sample is 32 men; treat the window resolution as the contribution and the point estimates as provisional.
- **Two mid-size Chinese time-series studies landed with tight intervals.** Nanning CVD mortality (87,913 deaths, 11 years) and Guangdong eczema outpatient visits (879,356 records) both report small but precisely bounded per-unit associations, with SO_2 carrying the largest single-pollutant coefficient in each. The eczema mixture analysis attributes 49% of the PM_{10}-based mixture effect to PM_{10} itself, but 38%/38% of the $PM_{2.5}$-based mixture to SO_2 and NO_2 — i.e. $PM_{2.5}$ mass is close to a passenger variable in that model.
- **South Asia is over-represented today, and by preprints.** Four of 18 records are South Asian, two of them Research Square preprints reporting single-station descriptive PM statistics (Cumilla, Bangladesh: February mean PM_{10} 210.7 $\mu g/m^3$). Neither is peer reviewed; neither reports instrument provenance or QA. Logged for coverage, not for citation.
- **Conflict as an exposure event was quantified.** Abi Nader et al. paired satellite and five-site ground $PM_{2.5}$ across the 2024 Beirut conflict; satellite median rose 15.11 $\rightarrow$ 20.21 $\mu g/m^3$, and BAPHE exposure–response functions give RR 1.011 respiratory and 1.032 skin admissions. The transferable part is that a ten-day handheld campaign can anchor a satellite comparison. **Trial watch: zero** ClinicalTrials.gov registrations or updates matched the window; the 23–29 July accumulation reports in the 2 August weekly rollup.

Corpus analytics

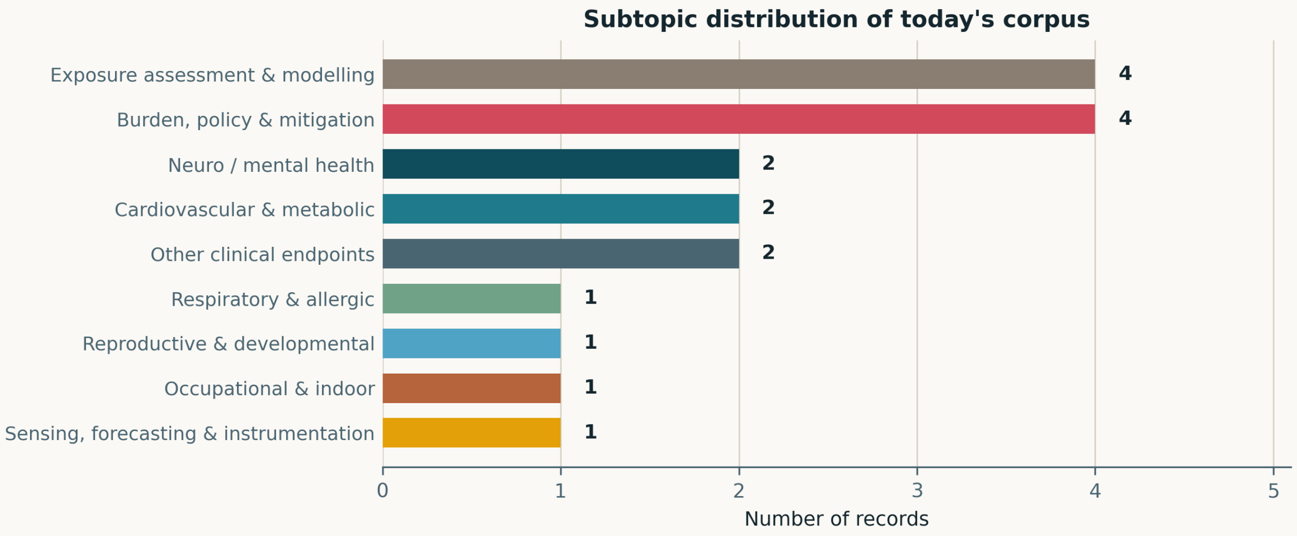
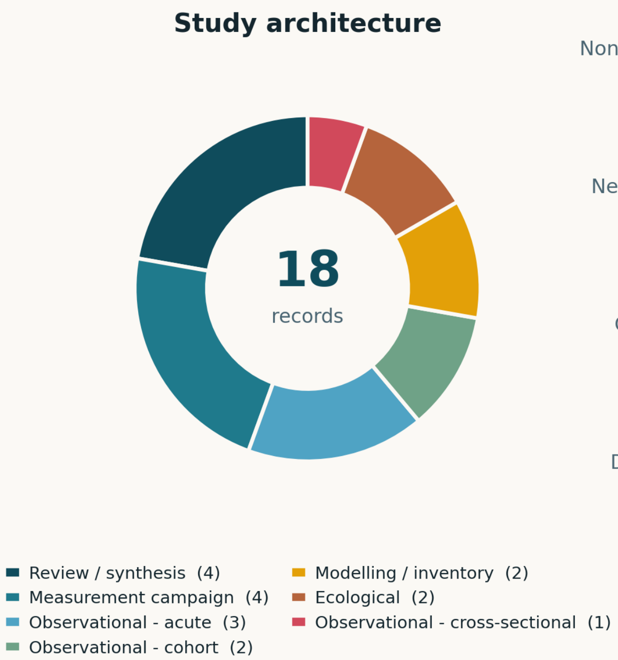


Fig. Subtopic assignment of the 18 in-scope records. Assignment is single-label by dominant contribution. *Exposure assessment & modelling* and *Burden, policy & mitigation* tie for the lead at four records each; *Mechanistic toxicology* is empty for the first time in this series, and *Sensing, forecasting & instrumentation* carries a single record — a commentary letter rather than primary work.



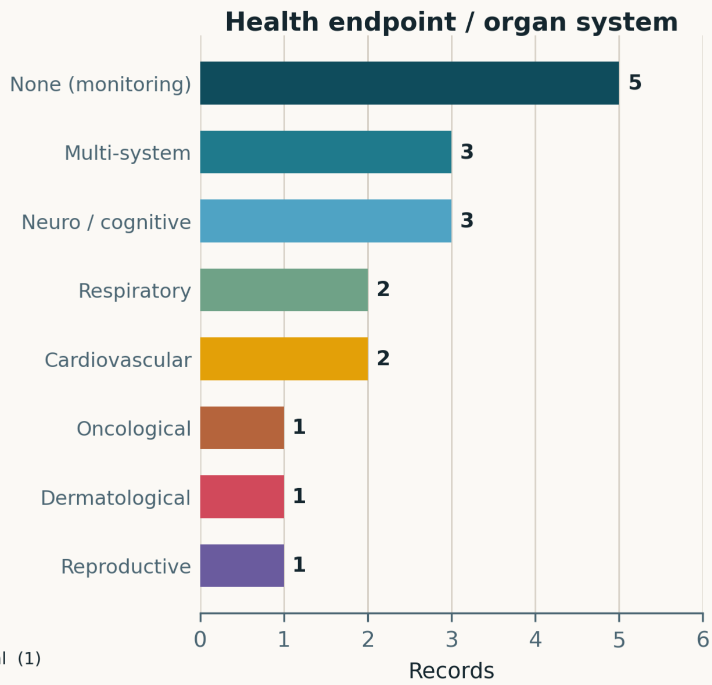


Fig. Left — study architecture. Reviews and syntheses account for 4/18 (22%) and measurement campaigns for a further 4; primary observational epidemiology (cohort, acute, cross-sectional) contributes 6. Right — health endpoint by organ system. Five records carry no human health endpoint at all — the four exposure-assessment records plus the forecasting commentary.

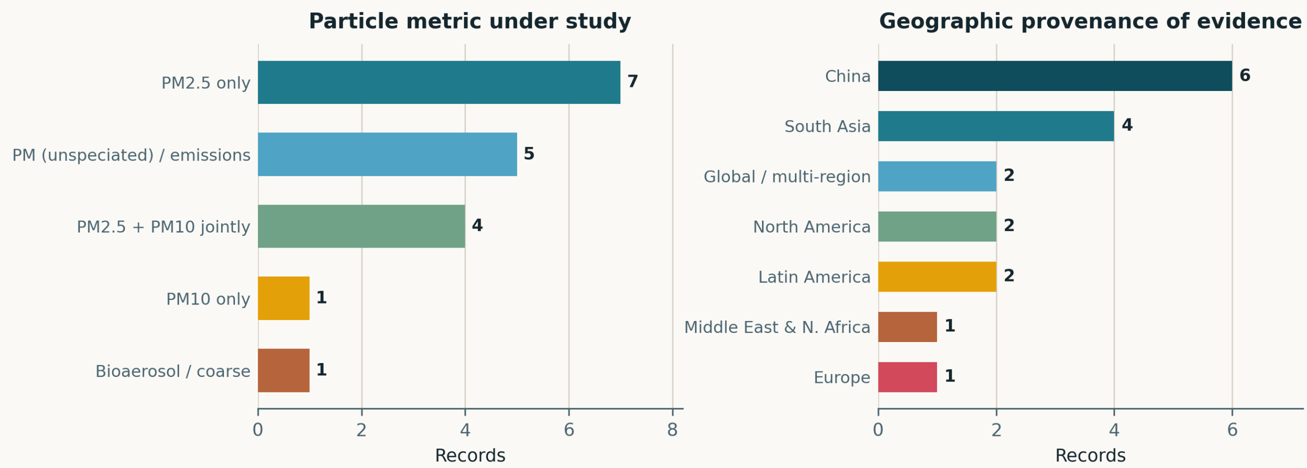


Fig. Left — particle metric. $PM_{2.5}$ alone is the exposure in 7 records and appears jointly with PM_{10} in a further 4; five records treat PM as unspeciated mass or as an emissions category, characteristic of the review and burden-estimate literature. **No record today addresses ultrafine particles** — the third consecutive issue in which the deposition-relevant size fraction is unrepresented. Right — geographic provenance. China contributes 6 records (33%), South Asia 4.

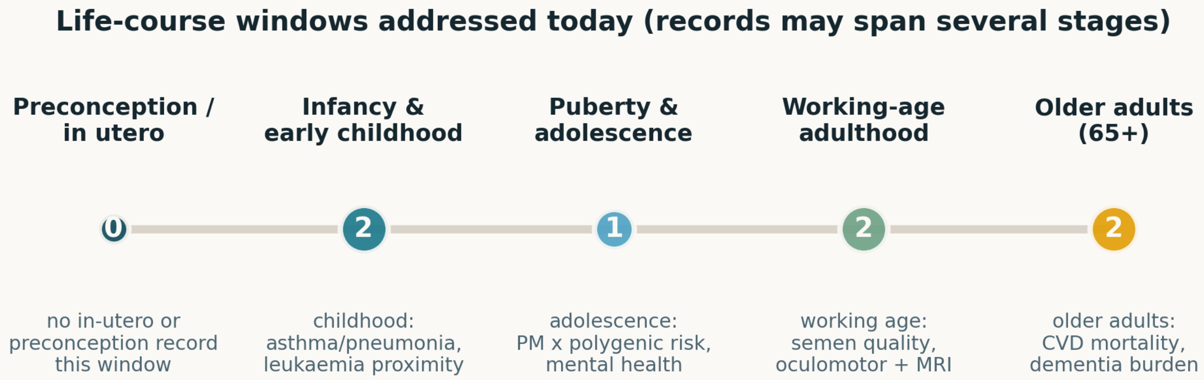


Fig. Life-course windows addressed. Counts are non-exclusive and cover only the seven records with an age-specific human population; the remaining eleven are monitoring, modelling, burden-estimate or review work with no defined life stage. **No record this window addresses preconception or in-utero exposure** — a break from the developmental-origins emphasis that dominated the 27–29 July issues.

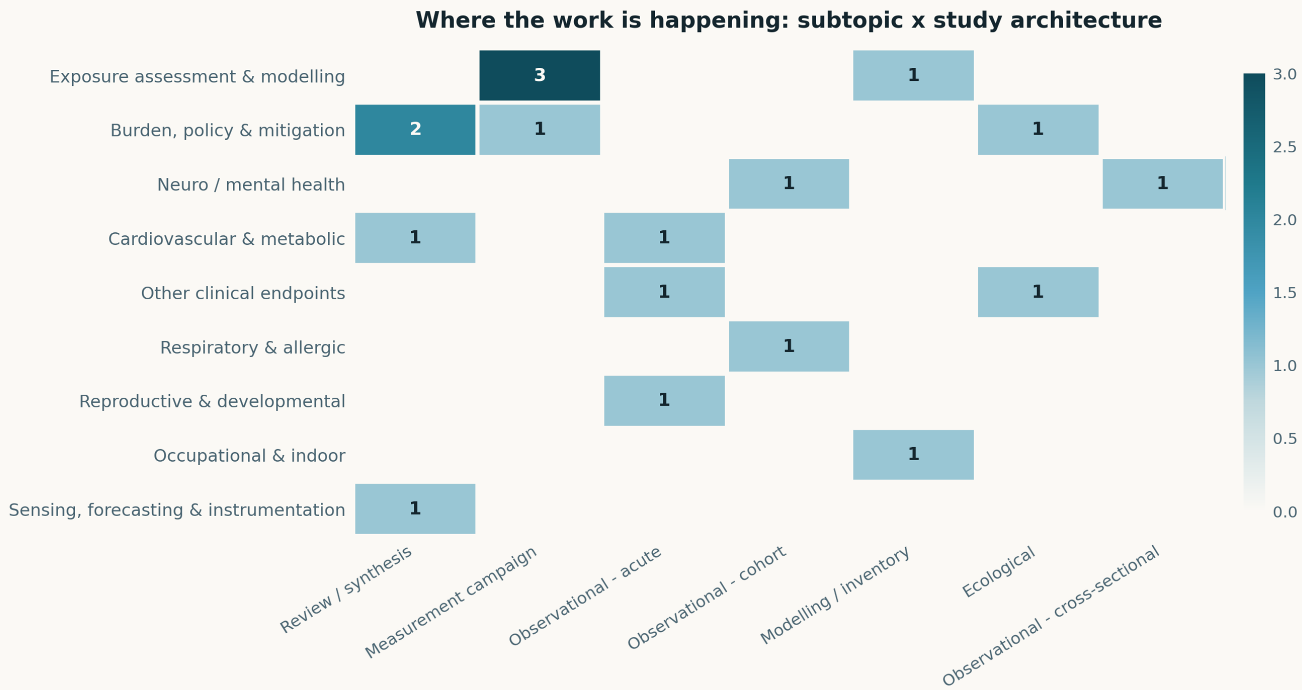


Fig. Subtopic × study-architecture cross-tabulation. The structural feature this series keeps flagging is starkest today: the *Exposure assessment* and *Sensing* rows carry no health-endpoint linkage at all, while every health-endpoint row is populated exclusively by designs that take exposure as a given input.

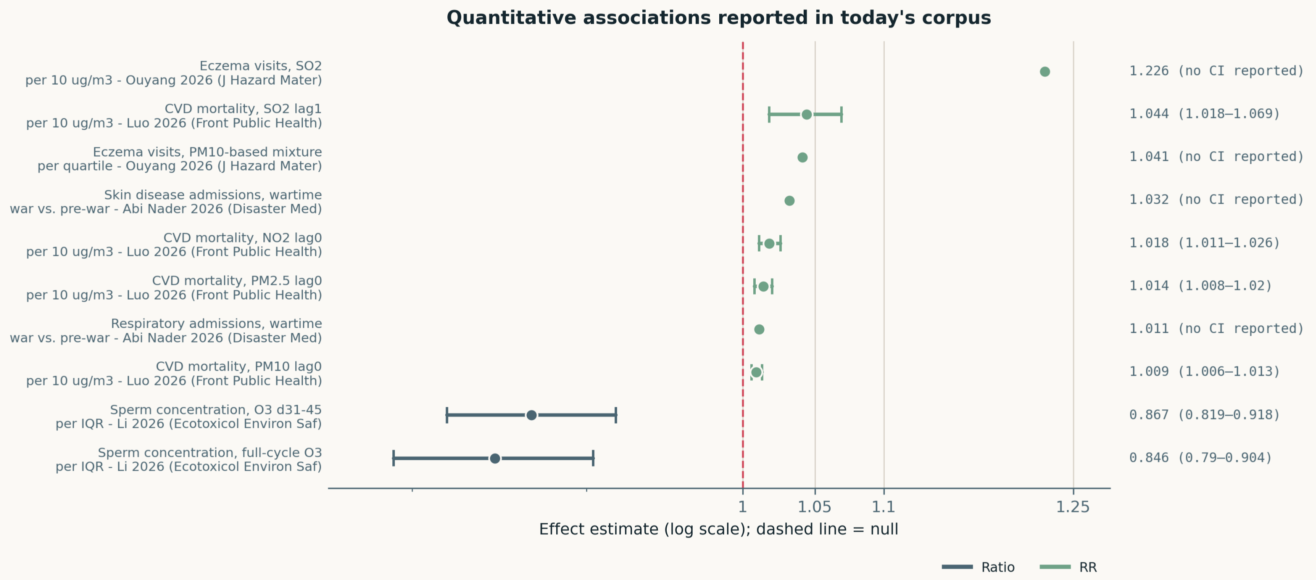


Fig. All quantitative effect estimates reported across the corpus, on a common logarithmic axis. **Exposure contrasts are heterogeneous and estimates are not mutually comparable**: Luo et al. report per-10 $\mu g/m^3$ single-day-lag RRs; Li et al. report per-IQR ratios, plotted below 1 because the association is a *reduction* in sperm concentration; Ouyang et al. mix per-10 $\mu g/m^3$ and per-quartile-of-mixture contrasts; Abi Nader et al. report a period contrast. Whiskers are drawn only where a confidence interval was reported — four of the ten estimates today carry none, which is itself a reporting-quality signal.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Exposure assessment & modelling 4 records

Zhang et al. 2026 npj Clim Atmos Sci

High-resolution aerosol liquid water content across the contiguous US using machine learning

ML surrogates trained on GEOS-Chem output to learn ALWC–predictor thermodynamics, then applied to bias-corrected high-resolution inputs, giving **1 km daily ALWC for 2000–2019**. Outperforms the parent CTM on daily variability and spatial heterogeneity. Regional ordering Midwest > East > West is driven by $PM_{2.5}$ mass, composition, temperature and RH; the secular decline tracks sulfate. Demonstrated as a constraint on water-soluble iron variability. The limitation is structural: the ML model inherits every bias of GEOS-Chem's thermodynamic module, so this is a downscaling of a simulation, not an observational product — there is no independent ALWC validation set. Still the most directly useful record here for hygroscopic-growth correction of optical sensors.

doi: 10.1038/s41612-026-01371-2

Chen et al. 2026 J Hazard Mater

Dual-isotope evidence for shipping contributions to atmospheric nitrate in coastal Chinese cities

615 daily $PM_{2.5}$ samples across Tianjin, Qingdao, Shanghai, Xiamen and Guangzhou during the cold-season transition, with nitrate δ^{15}N and δ^{18}O fed to a Bayesian isotope mixing model against an emissions-inventory prior. Shipping-derived nitrate ranged **17.6±4.7% (Tianjin) to 37.9±11.4% (Xiamen)**. Northern cities show shipping contribution rising with pollution severity and marine air-mass association; southern cities show it falling as inland transport intensifies. The inference threat is the mixing-model prior — source endmember δ^{15}N ranges for shipping overlap substantially with those for coal combustion, and no endmember sensitivity analysis is reported. Relevant to sensing because it quantifies how much of coastal $PM_{2.5}$ mass a mass-only monitor cannot attribute.

doi: 10.1016/j.jhazmat.2026.143129

Patwary 2026 Res Sq (preprint)

Spatio-temporal assessment of PM_{10} and meteorological correlates in Cumilla, Bangladesh

Single Department of Environment monitoring station, January–April 2026, Pearson correlation and trend analysis against temperature, RH and rainfall. Dry-season peak **210.7 μg/m^3 monthly mean PM_{10} in February**, roughly 4.7× the WHO 24-h guideline. No instrument model, calibration history or data-completeness statistics are reported, and one station cannot support the "spatio-temporal" framing in the title. Logged because South Asian monitoring records are structurally scarce in this corpus, not because the analysis is sound.

doi: 10.21203/rs.3.rs-10519411/v1

Raj & Dhada 2026 Res Sq (preprint)

Spatiotemporal ambient air quality across urban, tourist and industrial Himachal Pradesh

Five Himalayan towns (Solan, Baddi, Shimla, Kullu, Manali) contrasted by land use, 25 June–24 July 2026, 150 site-days, using AQI plus $PM_{2.5}$, PM_{10}, CO, SO_2, NO_2 and O_3 taken from the public AQI.in platform. The land-use contrast (industrial Baddi vs. tourist hill towns) is the design's one genuine idea. The fatal weakness is provenance: AQI.in aggregates heterogeneous and partly low-cost sources with no published calibration chain, so between-town differences cannot be separated from between-instrument differences. Preprint, not peer reviewed.

doi: 10.21203/rs.3.rs-10492149/v1

Burden, policy & mitigation 4 records

Abi Nader et al. 2026 Disaster Med Public Health Prep

Short-term respiratory and skin disease risks from air pollution during the 2024 Beirut conflict

Satellite-derived $PM_{2.5}$ over Beirut's southern suburbs compared pre-war (1 Aug–22 Sep 2024) against war (23 Sep–29 Nov 2024), plus ground $PM_{2.5}$ at five sites over ten days. Satellite median rose **15.11 → 20.21 μg/m^3**; ground median in the direct war zone 19.33 → 22.16 μg/m^3. Applying BAPHE exposure–response functions gives **RR 1.011 (AR 1.13%) respiratory** and **RR 1.032 (AR 3.10%) skin** admissions, corroborated by WHO AirQ+. The estimates are transferred, not observed — no admissions data were analysed — so these are projections conditional on a borrowed concentration–response function.

doi: 10.1017/dmp.2026.10412

Li et al. 2026b Ann Acad Med Singap

Burden trends of Alzheimer disease and other dementias in China, 1990–2023

GBD 2023 secondary analysis, GATHER-compliant. Deaths rose 122,411 → 586,002; DALYs and prevalence more than tripled; age-standardised rates rose only modestly, so the burden growth is demographic. Women carry higher ASRs in 2023 but male rates are rising faster; burden concentrates at ≥75 years. Among the modelled risk factors, **ambient particulate matter accounts for the largest attributable share in 2023**, while household solid-fuel air pollution declined. That attributable fraction is a GBD model output, not an epidemiological estimate from Chinese cohorts, and it inherits the comparative risk assessment's exposure–response assumptions wholesale.

doi: 10.47102/annals-acadmedsg.2026146

Vellaiyan 2026 J Environ Health Sci Eng

Diesel NO_x emissions and public health: a source–pathway–receptor synthesis

Review spanning combustion science, atmospheric chemistry, exposure assessment and epidemiology — disciplines that normally publish past each other. Useful on near-road gradients, street-canyon trapping, and the gap between certification cycles and real-world driving emissions. A synthesis, not new evidence; the epidemiological half is thinner than the combustion half.

doi: 10.1007/s40201-026-00995-z

Makkar et al. 2026 Environ Sci Pollut Res

Fungal biotechnology for air-pollution mitigation: mechanisms and engineered systems

Proposes mycofiltration as a complement to conventional filtration on the argument that HEPA-class media remove particulate mass efficiently but interact weakly with the chemically reactive fraction (aldehydes, PAHs, VOCs, transition metals). Mechanisms offered: extracellular oxidoreductases for aromatic VOC/PAH oxidation, chitin–glucan–melanin wall polymers for metal and soot immobilisation, porous mycelium

for interception. Laboratory and pilot-scale evidence only — no field deployment and no bioaerosol-emission safety data. Speculative.
doi: 10.1007/s11356-026-38029-8

Neuro, cognition & mental health — 2 records

Calderón-Garcidueñas et al. 2026 J Alzheimers Dis
Abnormal eye movements reflect early cortical and volumetric brain changes in $PM_{2.5}$-exposed urban youth
Video-based eye tracking (EyeLink 1000-plus) in 80 volunteers aged 33±11 from Metropolitan Mexico City and lower-exposure Cuernavaca, with structural MRI in 45 MMC subjects. **Oculomotor variables did not differ significantly between cities** — the headline null. Abnormal antisaccades, low pursuit gain and square-wave jerks were merely *more frequent* in MMC, and the significant findings are correlations between oculomotor measures and frontal–temporal–parietal cortical, hippocampal, thalamic, caudate, amygdalar and cerebellar structure *within* the exposed group. With $n = 45$ and that many regions the multiple-comparison exposure is severe, and no correction is described. The group's long-running MMC series is valuable, but this instalment does not establish an exposure contrast.
doi: 10.1177/13872877261471048

Cattarinussi et al. 2026 medRxiv
Independent and joint effects of outdoor air pollution and genetic risk on adolescent mental health trajectories
ABCD Study, $n = 10{,}620$, ages 9–16, sex-stratified trajectories of internalising symptoms and psychotic-like experiences against residential $PM_{2.5}$ and polygenic scores for schizophrenia and major depression. $PM_{2.5}$ was associated with **smaller decreases in PLEs over time in females (p-FDR = 0.005)**, with no effect on internalising trajectories in either sex, and a **$PM_{2.5}$ × PRS-MDD interaction on PLE trajectories in females (p-FDR = 0.048)** — a borderline interaction that will not survive replication unless the effect is real. $PM_{2.5}$ is residential-address-assigned, so within-person exposure misclassification is large and non-differential, which biases toward the null for main effects but has unpredictable sign for interactions. Preprint.
doi: 10.64898/2026.07.12.26357864

Cardiovascular & metabolic — 2 records

Luo et al. 2026 Front Public Health
Short-term ambient air pollution and cardiovascular mortality: an 11-year time-series in Nanning
87,913 CVD deaths, 2014–2024; generalized additive quasi-Poisson adjusted for long-term trend, temperature, RH and day of week, across single-day lags 0–7 and cumulative windows 0–1 to 0–7. At lag 0, **RR 1.014 (1.008–1.020) for $PM_{2.5}$**, 1.009 (1.006–1.013) for PM_{10} and 1.018 (1.011–1.026) for NO_2; SO_2 peaks at lag 1 with **1.044 (1.018–1.069)**, the largest coefficient in the set. Subtropical southwestern China is genuinely under-represented in this literature. Two-pollutant models were run, but the paper does not resolve the $SO_2/PM_{2.5}$ collinearity its own results imply, so the apparent SO_2 dominance should not be read as source-specific.
doi: 10.3389/fpubh.2026.1878398

Chrysohoou et al. 2026 Eur J Prev Cardiol
Preventing cardiovascular disease within the built and living environment: an expert opinion
Expert opinion placing air and noise pollution alongside food environment, green space, physical activity and psychosocial stress in a single built/living/social framework, with inflammation, oxidative stress, neuroendocrine dysregulation and chronic stress as the shared mechanistic substrate. No new data and no quantitative synthesis; its value is as a citable framing document for grant and introduction text, nothing more.
doi: 10.1093/eurjpc/zwag387

Other clinical endpoints — 2 records

Ouyang et al. 2026 J Hazard Mater
Independent and joint effects of short-term air pollutant exposure on eczema in two Chinese cities
879,356 daily eczema outpatient records from two Guangdong dermatology hospitals matched to daily $PM_{2.5}$, PM_{10}, NO_2, SO_2 and O_3; distributed lag non-linear models for single pollutants and quantile g-computation for two four-pollutant mixtures. Per $10\,\mu g/m^3$, **SO_2 gave the largest single-pollutant effect at +22.63%**. Each quartile increase in the PM_{10}-based mixture raised visits 4.14% (49.5% of the weight on PM_{10}); the $PM_{2.5}$-based mixture raised them 4.21%, but with the weight on SO_2 (38.1%) and NO_2 (37.5%). Both mixtures correspond to ≈18 extra visits per day. Outpatient visits confound severity with care-seeking behaviour, and no confidence intervals are reported for the mixture estimates.
doi: 10.1016/j.jhazmat.2026.143124

Zurita et al. 2026 Pediatr Blood Cancer
Childhood leukaemia and proximity to benzene emission sources in the Mexico City area
Ecological study of acute lymphoblastic leukaemia cases from one paediatric hospital, 2010–2021, against counts of benzene-emitting establishments (petroleum and coal product manufacture, fuel and lubricant retail, printing, chemical industry) within distance buffers, using multivariate spatial models by municipality, sex and age group. Significant positive association with source counts for ages 2–17. Single-hospital case ascertainment plus municipality-level exposure is a textbook ecological design: no individual exposure, no residential history, and referral patterns to one hospital are almost certainly spatially structured. Hypothesis-generating at best — the authors say as much.
doi: 10.1002/1545-5017.70482

Respiratory & allergic — 1 record

Wang et al. 2026 Front Med
Environmental and socioeconomic influences on childhood asthma and pneumonia burden
Retrospective cohort from paediatric electronic medical records at one Wuhu hospital, Anhui, January 2022–December 2025: 936 children aged 1–14 (47.4% asthma, 36.5% pneumonia, 16.0% both), with regional NO_2, $PM_{2.5}$, temperature and humidity and census-based socioeconomic status. Higher NO_2 and $PM_{2.5}$ were associated with asthma; biomass fuel use and poor household ventilation with pneumonia; lower SES strongly with pneumonia but not asthma. Pneumonia peaked in winter, asthma in spring. Hospital-based case series with area-level exposure and no comparison population — the associations describe who among the diagnosed had which condition, not incidence.

doi: 10.3389/fmed.2026.1872723

Reproductive & developmental 1 record

Li et al. 2026 Ecotoxicol Environ Saf

Ambient air pollutant exposure and semen quality: a refined multi-window panel study

One-year prospective panel of 32 male undergraduates aged 18–22 with four quarterly follow-ups spanning the 90-day spermatogenic cycle, using both daily-average and cumulative-dose exposure metrics and — the methodological contribution — stratifying the 15–69-day meiotic window into three biologically distinct sub-stages. Linear mixed-effects, two-pollutant and Bayesian weighted quantile sum models. Per IQR of full-cycle O_3, **sperm concentration fell 15.45% (95% CI −20.95 to −9.58)**, with the sub-stage peak at days 31–45 (−13.30%, −18.09 to −8.24); NO_2 acted earliest (15–30 d), CO later (31–69 d). With $n = 32$ the sub-stage resolution outruns the power, and homogeneous undergraduates limit generalisability by design. The window-stratification framework is worth borrowing regardless.

doi: 10.1016/j.ecoenv.2026.120581

Occupational & indoor exposure 1 record

Gwak et al. 2026 Risk Anal

Size-resolved modelling of airborne viral RNA copies in indoor environments

Size-resolved framework over 1–2000 μm coupling saliva-specific evaporation and residue formation with gravitational settling, across temperature, RH and two ambient pressures (101.325 and 75.3 kPa), combined with saliva-mass uncertainty, viral load, half-life, exhalation mode and ventilation. **Talking dominates airborne RNA persistence** because coughing and sneezing generate larger droplets that settle quickly. Predictions matched hospital air-sampling data within roughly one order of magnitude at the 50th saliva-mass percentile. The framework's own caveat is the binding one: RNA decay stands in for infectivity decay, and if viral RNA preferentially concentrates in long-suspended fine aerosol the size-resolved conclusions shift. Directly relevant to size-resolved optical sensing of bioaerosol in occupied spaces.

doi: 10.1111/risa.70318

Sensing, forecasting & instrumentation 1 record

Aher & Ravindran 2026 Environ Sci Pollut Res

Comment on advancing air pollution forecasting: operational challenges beyond predictive accuracy

Letter responding to a 2026 forecasting review. The substantive critique is that the review optimises for predictive accuracy while under-treating what actually breaks operational systems: model transferability into data-sparse regions, uncertainty quantification, interpretability of deep architectures, and input data-quality constraints. It points to adaptive mesh refinement, AI-assisted chemical transport models and hybrid physics–ML frameworks as the productive direction. No new results — but the transferability-into-data-sparse-regions argument is precisely the problem low-cost sensor networks exist to solve, and the letter states it cleanly.

doi: 10.1007/s11356-026-38087-y

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Calderon-Garcidueñas	Abnormal eye movements reflect early cortical and volumetric brain changes in $PM_{2.5}$-exposed urban youth	*J Alzheimers Dis*	Cross-sectional imaging	$PM_{2.5}$	Mexico
Cattarinussi	Independent and joint effects of outdoor air pollution and genetic risk on adolescent mental health trajectories	*medRxiv*	Prospective cohort	$PM_{2.5}$	USA
Cardiovascular & metabolic					
Chrysohoou	Preventing cardiovascular disease within the built and living environment: an expert opinion	*Eur J Prev Cardiol*	Narrative review	PM (unspeciated)	EU-27
Luo	Short-term ambient air pollution and cardiovascular mortality: an 11-year time-series in Nanning	*Front Public Health*	Time-series	$PM_{2.5}$ + PM_{10}	China
Reproductive & developmental					
Li	Ambient air pollutant exposure and semen quality: a refined multi-window panel study	*Ecotoxicol Environ Saf*	Panel study	$PM_{2.5}$ + PM_{10}	China
Respiratory & allergic					
Wang	Environmental and socioeconomic influences on childhood asthma and pneumonia burden	*Front Med*	Retrospective cohort	$PM_{2.5}$	China
Occupational & indoor					
Gwak	Size-resolved modelling of airborne viral RNA copies in indoor environments	*Risk Anal*	Physical exposure model	Bioaerosol / coarse	Global
Sensing, forecasting & instrumentation					
Aher	Comment on advancing air pollution forecasting: operational challenges beyond predictive accuracy	*Environ Sci Pollut Res*	Commentary / methods letter	PM (unspeciated)	India
Exposure assessment & modelling					
Chen	Dual-isotope evidence for shipping contributions to atmospheric nitrate in coastal Chinese cities	*J Hazard Mater*	Source apportionment	$PM_{2.5}$	China
Patwary	Spatio-temporal assessment of PM_{10} and meteorological correlates in Cumilla, Bangladesh	*Res Sq (preprint)*	Field measurement	PM_{10}	Bangladesh
Raj	Spatiotemporal ambient air quality across urban, tourist and industrial Himachal Pradesh	*Res Sq (preprint)*	Field measurement	$PM_{2.5}$ + PM_{10}	India
Zhang	High-resolution aerosol liquid water content across the contiguous US using machine learning	*npj Clim Atmos Sci*	Machine-learning model	$PM_{2.5}$	USA
Burden, policy & mitigation					
Abi Nader	Short-term respiratory and skin disease risks from air pollution during the 2024 Beirut conflict	*Disaster Med Public Health Prep*	Environmental surveillance campaign	$PM_{2.5}$	Lebanon
Li	Burden trends of Alzheimer disease and other dementias in China, 1990-2023	*Ann Acad Med Singap*	GBD secondary analysis	PM (unspeciated)	China
Makkar	Fungal biotechnology for air-pollution mitigation: mechanisms and engineered systems	*Environ Sci Pollut Res*	Critical review	$PM_{2.5}$	India
Vellaiyan	Diesel NOx emissions and public health: source-pathway-receptor synthesis	*J Environ Health Sci Eng*	Narrative review	PM (unspeciated)	Global
Other clinical endpoints					
Ouyang	Independent and joint effects of short-term air pollutant exposure on eczema in two Chinese cities	*J Hazard Mater*	Time-series	$PM_{2.5}$ + PM_{10}	China
Zurita	Childhood leukaemia and proximity to benzene emission sources in the Mexico City area	*Pediatr Blood Cancer*	Ecological panel	PM (unspeciated)	Mexico

Provenance and method

Entry window **30 July 2026**, contiguous with the 29 July issue — no gap. Sources: **PubMed** (E-utilities and connector) queried on the `[Date - Entry]` field along two independent axes, health (16 hits) and instrumentation (66 hits); **Europe PMC** `CREATION_DATE:[2026-07-30 TO 2026-07-30]` (5 hits, 3 of them duplicates of PubMed records, 2 new preprints); **arXiv** relevance query with a local recency screen (60 entries, none newer than 29 July, none in scope); **OpenAlex** returned HTTP 429 and remains unusable for date-windowed harvesting; **ClinicalTrials.gov** returned zero registrations or updates in the window. **18 records in scope.**

Two rejection groups deserve explicit statement. (i) The 66-hit instrumentation axis was almost entirely recall noise — `calibration`, `machine learning` and `monitoring` match clinical prediction-model papers — and yielded two in-scope records. (ii) The **Consensus** sweep, which exists precisely because PubMed does not index *Atmos. Meas. Tech.* or *Atmos. Chem. Phys., has no entry-date filter*. Its eight low-cost-sensor hits resolved via Crossref to publication dates between 20 January and 6 July 2026, and one (`10.1080/02786826.2026.2676293`) was already carried in the 29 July issue. All eight were rejected to `state/rejected.jsonl` rather than presented as new. The structural consequence is that **this pipeline currently has no date-windowed channel into the atmospheric-measurement literature**; closing it needs a Crossref by-journal feed for AMT/ACP keyed on `created`, not another Consensus query.

Subtopic, design, particle-metric and geography labels were assigned from title, abstract, keywords and MeSH terms; assignment is single-label by dominant contribution and is therefore a judgement, not a controlled vocabulary. Effect estimates in the forest plot are transcribed verbatim from abstracts and use heterogeneous exposure contrasts — see the figure caption. Two records are Research Square preprints and one is a medRxiv preprint; none has been peer reviewed, and each is marked as such in the digest. All DOI links resolve to the publisher of record. Content derived from PubMed and Europe PMC metadata; abstracts remain the copyright of their respective publishers.